Anonymisation Report - Group V

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November 2026

1 What we did

The survey contained 200 records of people with their name, sex, date of birth, ZIP code, education, citizenship, marital status, voting method (whether the vote was cast electronically or in-person), and party preference. We also got a public register of 1495 residents, and the published election results.

We aimed to make it difficult to de-anonymize people in the survey using the data in the public register. If someone were to match an anonymous record in the survey with a named person in the public register using (combinations of) quasi-identifiers such as date of birth and ZIP code, they could find out someone's political views. Therefore, we made the quasi-identifiers less precise to protect against such data linkage, while retaining insightfulness so that statistical tests can still be conducted.

2 Changes to the data

The alterations we did to the original survey are as follows:

- **Name**s were removed, as they directly identify people. This should not be said, but it has to be because this is a declaration of changes made to the dataset and it has to stay accurate.
- **Citizenship** was removed due to a high risk of reidentification. Approximately 93% of the people in the dataset had Danish citizenship, while other nationalities were sparsely represented; thus making it very easy to reidentify non-Danish citizens based on nationality alone. We

tried to convert the Citizenship variable to a binary Danish/non-Danish variable, but even then, there was a significant reidentification risk. As such, we decided to drop this variable entirely.

- **Date of Birth** was transformed into **Age brackets** for better k-anonymity. Ages are grouped into 10 year bins: 20-29, 30-39, 40-49, 50-59, 60-69, 70+.
- **Married status** was converted into a binary variable (Married / Not Married).
- **Education** was merged into two groups, Higher Education and Other Education. One of the original groups (Bachelors programmes) had a single person, which is easy to spot for anyone who knows the person.

The other variables - **Sex**, **ZIP**, **Party**, and **Evote** - are unchanged.

3 Disclosure Risk Assessment

The main privacy risk associated with the data from the survey was the data linkage. Even with the direct identifiers removed, there is a potential risk of identifying an individual by cross referencing survey data with external sources. An attacker, who has access to both survey data and the public registry, could potentially find out voters' information with help of the public registry.

To evaluate and contain this risk, we have applied generalization to the quasi-identifiers, removed the direct identifier and the citizenship, and assessed the resulting data set using k-anonymity. The assessment covered the four attributes, sex,age,zip code and marital status, which were common between the survey and the public registry. The measured k value represents the size of the voters' equivalence class, meaning how many people in the registry share the same values. Higher k value means that a person is "hidden" among more people with the same characteristics.

The results for k-anonymity indicated that the anonymization was mostly successful. None of the 200 individual persons from the survey matched exactly one person in the registry. The smallest equivalent class had a k value of 2 and the average group size was 22.5. Four persons from the survey belonged to groups of 5 or less persons, and 16 persons from the survey belonged to groups of 10 or fewer persons. The calculated average probability of correctly identifying a person from the survey was ca. 6% (mean of the 1/k across all respondents).

The results suggest that the generalization of the quasi-identifiers, in combination with the removal of the direct identifiers and the citizenship status, has reduced the risk of direct linkage significantly. The minimum k value of 2 indicates that, an attacker who has prior information about relevant attributes could at best narrow down the identity down to 2 possible individuals. We have also looked in to the uniqueness measures within the survey itself. By using the same four attributes that shared between the public registry and the survey, the 200 individuals were divided in 70 groups with 17 individuals being unique (having a unique combination of those four attributes) in the survey. This indicates that a combination of demographic attribute can isolate an individual even after the generalization. But, the uniqueness of the survey does not imply that an individual can be identified by the name, an attacker would still require additional information. When education level was added to the shared attributes, the number of unique

individuals in the survey increased to 59. Which indicates that education level attribute can increase the identification risk even though the education level is not included in the public registry.

3.1 Attribute Disclosure

In addition to data linkage, we have also assessed the attribute disclosure risk. Political party preference is the sensitive attribute in this dataset. Since a group containing a single individual is inherently homogenous, we have only considered the groups which contain a minimum of 2 individuals. In the assessment, we have identified 19 unique groups, where all the individuals in each group have voted for the same party. All together, these 19 groups contained 57 individuals, where only 16 of them belonged to groups consisting of 5 or more individuals. This indicates that, while the anonymization prevents unique matching against the entries in public registry, it will not fully prevent an attacker from inferring political preferences.

Reducing the attack surface for this form of attribute disclosure would require further generalization or suppression. However, considering the small sample size of 200 individuals in the survey, further generalization could reduce the analytical value of the dataset. Therefore we have decided to accept the remaining risk as a trade off between data protection and the quality of the data. The reported k values should be considered as the best case, rather than guarantees of anonymity against all possible attacks, given that attackers who have more information than our original assumption, could infer more individuals' identities.

In general, we have identified that the anonymization has reduced the possibility of linking individuals in the survey data directly to the persons in the public registry. There are no unique matches and the average equivalence class is relatively large. However, the small groups identified in the analysis and the presence of the homogeneous groups with party preference indicates that there are some residual disclosure risks, which we have identified as the main limitation of the current approach.

4 Analyses - before and after

Before and after the transformations for anonymization, we did the three analyses. As ZIP, Sex, Party, and Evote weren't changed, statistics involving these didn't change either.

4.1 (A) Political preferences in the survey vs. election results by voting method

No significant difference was found in the distribution of political preference for either voting channel (e-vote or in-person) when comparing the anonymized survey to the public election results. Because neither the party preferences nor the e-vote statistics were changed during anonymization, this is identical to our analysis on the raw survey data as well.

4.2 (B) Party preference vs. demographics

We tested whether there is a correlation between the identity and background of the people in the survey (Age, Sex, Education, Marital status, ZIP code, and Citizenship) and their political preferences. The results are illustrated in Table 1.

Table 1: Significance of correlation between various identifiers and party preference, $\alpha = 0.05$

Attribute	**Raw data**	**Anonymised**
Age	$p = 0.0015$, significant	$p = 0.067$, not significant
Sex	$p = 0.0001$, significant	same
Education	$p = 0.0019$, significant	$p < 0.0001$, significant
Marital status	$p = 0.57$, not significant	$p = 0.75$, not significant
Zip	$p = 0.55$, not significant	same
Citizenship	$p = 0.97$, not significant	dropped

4.3 (C) Voting channel vs. demographics

Before transforming the data, age was the only variable which had a significant correlation with the method of voting (young people voted electronically more frequently). After binning the ages, age ceased to be significant.

5 Conclusions

Our main goal was to keep as much of the information in the survey as possible while still making it hard to link people to the public register. We think the result is a good balance.

The main findings survived the anonymization. The party preferences in the survey still match the election results, and sex and education still clearly correlate with party preference. The only thing we lost is the significance of age, and only by a little for party preference and voting channel. The trends are still visible in the data: older people lean more towards the Red party, and younger people vote online more often. We think this loss is totally acceptable.

On the privacy side, nobody in the survey can be matched to a single person in the register, and the average chance of picking the right person is around 6%. Still, the risk is not zero, and we only measured it against the register and this report. If an attacker has more information than we assumed, our dataset would be weaker than these numbers suggest. If we did this again, we would anonymize more aggressively, for example with larger age bins (15 years instead of 10), by suppressing the smallest groups or drop some of the attributes, even if that costs a bit more utility. We would rather lose a little more of the analysis than make de-anonymization easy for an attacker with extra knowledge.